Evaluating Semantic Automata on the Verification of Iterated Quantifiers

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Do English Iterated Quantifiers place demands on cognitive resources as predicted by Semantic Automata Theory?

Quantifier Verification

- Generalized Quantifiers[2]
  - Noun phrases that denote a set of sets
- Iterated Quantifiers
  - Sentences that contain multiple quantifiers
- Verification Strategies[2]
  - Cognitive algorithms used to determine the truth-value of a quantified sentence

Semantic Automata[2]

- Associates quantifiers with algorithmic procedures
  - ie. pushdown or finite state automata
- Makes specific predictions of the cognitive effort involved with quantifier verification[1]
  - e.g., by associating quantifiers with procedures and amount/type of working memory recruited

COMPLEXITY
- Aristotelian (all, no, some)
- Numeric (at least *n*, at most *n*)
- Parity (an even number of *n*)
- Proportional (most, more)

In This Study:

- Experimental evidence for SA in single quantifiers (Bremnes, et al. 2022; a.o.)
- SA makes predictions for iterated quantifiers
- Partial evidence: Szymanik et al. (2012) in Polish

Predictions

1. True instances of Every-Some more complex to verify (i.e. lower accuracy, longer VTs) than True instances of Some-Every, and vice-versa for False instances
2. Iterated and proportional quantifiers will both actively recruit working memory
3. Aristotelian quantifiers will be less complex to verify than iterated and proportional
4. Exploratory: Quantifier Encoding. Reading times at the 1st occurrence of a quantifier faster than at the 2nd.

Current Study

Procedure

[1] Ishihara Test **[2] Digit Span Task** **[3] Arrow Flanker Task**

[4] Sentence-Picture Verification Task
- Moving-window self-paced reading task
  - To explore encoding differences between quantifier position
- Followed by an image either satisfying the sentence or not.
- Offline Task: determine truth-value of the sentence wrt the image.
- Dependent measures: verification time and accuracy.
- 40 total stimuli
  - 32 target items (8 x 4 Quantifier Conditions)
  - 8 attention checks

Stimuli Design

Class	Quantifier	T-Value	Example Sentence
Iterated	Every-Some	T/F	Every circle is connected to some square in the house.[Fig. 1]
Iterated	Some-Every	T/F	Some triangle is connected to every star in the scroll.[Fig. 2]
Aristotelian	All	T/F	All stars are pink in the heart.[Fig. 3]
Proportional	Most	T/F	Most squares are orange in the triangle.[Fig. 4]
Attention Checks	Existential	T/F	There are brown triangles in the star.

Figure 1. Iterated

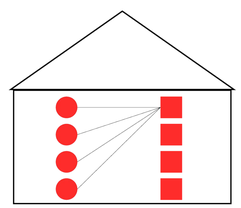


Figure 2. Iterated

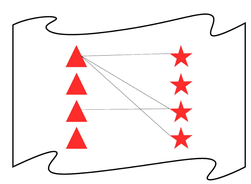


Figure 3. Aristotelian

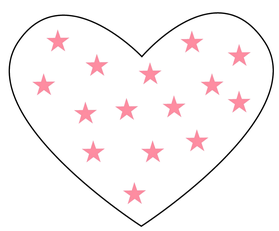
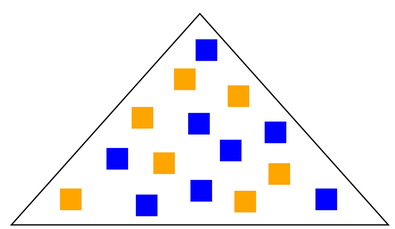


Figure 4. Proportional



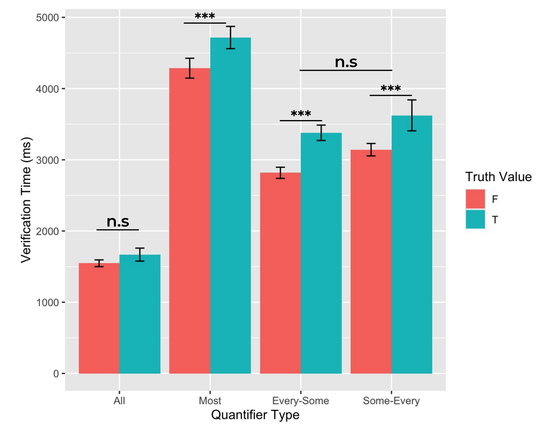
Selected References

[1] Szymanik, J., et al. (2012). *Logic and Interaction RAtionality Yearbook 2012.* [2] van Benthem, J. (1986). *Essays in logical semantics.* [3] Stoet, G. (2010). *Behavior Research Methods.* [4] Stoet, G. (2017). *Teaching of Psychology.*

Results

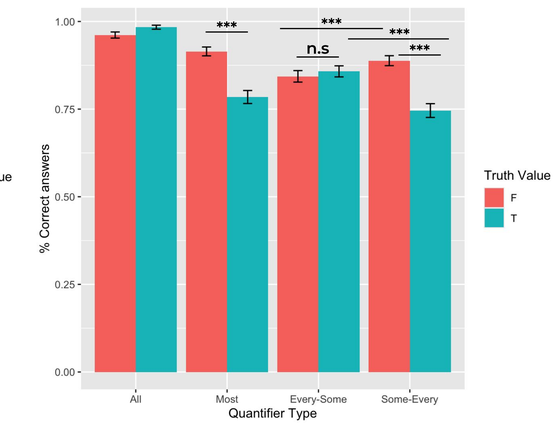
Participants: L1 English-speaking adults with normal/corrected vision recruited via Prolific completed a web-based experiment

Verification Time



- Significant effect of Quantifier and Quantifier/Truth-Value Interaction:
  - Fastest: Aristotelian
  - Slowest: Proportional
- Proportional and Iterated quantifiers were **significantly slower in their True condition**
- **No significant difference between Every-Some/Some-Every by Truth Value**
- Positive Quantifier/Digit Span Interaction
- Positive Quantifier/Truth Value and both Cognitive Tasks Interaction

Accuracy



- Significant effects of Quantifier, and Quantifier/Truth Value Interaction
- Significantly **higher accuracy for Aristotelian vs. all other quantifiers**
  - No significant differences between Proportional and Iterated
- **Significantly higher accuracy** in True instances of Every-Some than Some-Every
  - **No difference** between their False instances
- Positive effect of Digit Span
- **Significant** positive effect of the between **Quantifier/Digit Span** interaction for Proportional and Iterated quantifiers

Predictions	Szymanik, et al. (2013)	Viets & De Santo
1	≈	✘
2	✔	✔
3	Not tested	✔
4	Not tested	✘

Discussion

- Only partial support for SA
  - Compare to other algorithmic theories of quantifier verification (e.g. Bott, et al. 2019)
- Further study of the relation between Iterated quantifiers and properties of the scene (Feigenson, et al. 2004)

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